

Knowledge Test

P-01.

What is the fundamental difference between proportional and non-proportional reinsurance?

Conceptual

- A. In proportional reinsurance, the reinsurer covers losses above a retention; in non-proportional reinsurance, it shares premiums and losses in the same proportion.
- B. In proportional reinsurance, premiums and losses are shared in the same proportion; in non-proportional reinsurance, the reinsurer covers only the portion of the loss that exceeds the cedent's retention.
- C. Proportional reinsurance applies only to personal insurance risks; non-proportional reinsurance applies only to property damage risks.
- D. They produce equivalent economic results; the difference is purely administrative.

Answer P-01.

Proportional vs. Non-Proportional

Correct answer: B

Proportional reinsurance: Premiums and losses are shared between the cedent and the reinsurer in the *same proportion* specified in the contract. If the cedent cedes 60% of the premium, the reinsurer pays 60% of each loss. Types: quota share and surplus.

Non-proportional reinsurance: The reinsurer responds only when the loss (or aggregate loss experience) exceeds a predefined threshold (retention). Types: per-risk XL, per-occurrence or catastrophe XL, and aggregate stop-loss.

Proportional:

$$\text{Payment}_{\text{reins}} = \alpha \times \text{Loss}, \quad \alpha = \text{fixed ceded percentage.}$$

Non-proportional:

$$\text{Payment}_{\text{reins}} = \min(\max(\text{Loss} - M, 0), L).$$

Here, M is the retention (priority) and L is the coverage limit.

Key Point for HDI

Hannover Re is the group's natural reinsurance partner. In practice, HDI Colombia combines both types: proportional treaty reinsurance to make portfolio exposure more homogeneous, and catastrophe XL to protect regulatory capital against extreme events (earthquakes and heavy rainfall in cities).

P-02.

A per-occurrence XL contract has a retention of $M = 500$ million COP and a limit of $L = 4,000$ million COP. A loss of 3,200 million occurs. How much does the reinsurer pay?

Calculation

- A. 2,700 million COP
- B. 3,200 million COP
- C. 4,000 million COP
- D. 500 million COP

Answer P-02.**Payment Under a Per-Occurrence XL Contract**

Correct answer: A — 2,700 million COP

Expressing amounts in millions of COP:

$$\begin{aligned}\text{Payment}_{\text{reins}} &= \min(\max(\text{Loss} - M, 0), L) \\ &= \min(\max(3,200 - 500, 0), 4,000) \\ &= \min(2,700, 4,000) \\ &= 2,700.\end{aligned}$$

The cedent retains 500 million COP and the reinsurer pays 2,700 million COP. The layer covered by the contract extends up to $500 + 4,000 = 4,500$ million COP. Since $3,200 < 4,500$, the loss falls within this layer.

If the Loss Had Been 5,000 Million COP

$$\begin{aligned}\text{Payment}_{\text{reins}} &= \min(5,000 - 500, 4,000) \\ &= \min(4,500, 4,000) \\ &= 4,000.\end{aligned}$$

The cedent absorbs the excess:

$$5,000 - 500 - 4,000 = 500 \text{ million COP in additional losses.}$$

These 500 million are added to the initial retention of 500 million COP. A second XL layer would be required to cover that excess.

Standard Notation

The contract is written as **4,000 xs 500**: limit **xs** retention. The cedent bears the retention; where multiple layers exist, coverage is structured in layers.

P-03.

Under a surplus treaty, a policy has a sum insured (SI) of 10,000 M, a net retention of 2,000 M, and 4 ceded lines. What percentage of the loss does the reinsurer bear?

Calculation

- A. 40%
- B. 60%
- C. 80%
- D. 20%

Answer P-03.

Surplus Treaty: Ceded Percentage

Answer: C — 80%

Under a surplus treaty, the cedent retains its **retained line** (net retention) and cedes the excess up to N lines, where one line equals the retention.

Expressing amounts in millions:

$$\begin{aligned}\text{Net retention} &= 2,000, \\ \text{Ceded lines} &= 4, \\ \text{Ceded capacity} &= 4 \times 2,000 = 8,000, \\ \text{Total SI} &= 2,000 + 8,000 = 10,000.\end{aligned}$$

Therefore:

$$\begin{aligned}\text{Percentage ceded to the reinsurer} &= \frac{8,000}{10,000} \times 100\% = 80\%, \\ \text{Percentage retained by the cedent} &= \frac{2,000}{10,000} \times 100\% = 20\%.\end{aligned}$$

For a loss of amount X :

$$\text{Payment}_{\text{cedent}} = 0.20 \times X, \quad \text{Payment}_{\text{reins}} = 0.80 \times X.$$

Unlike quota share, surplus reinsurance is *selective by policy*: only policies exceeding the retention are ceded, allowing the cedent to retain small, homogeneous risks in full.

Advantage of Surplus Reinsurance

Reduced reinsurance commission on small risks (the cedent does not cede unnecessary premium), along with greater capacity for large risks without changing the retention by line of business.

P-04.

What does the *Rate On Line* (ROL) measure in XL reinsurance, and how does it relate to the return period?

Conceptual

- A. $ROL = XL \text{ premium} / \text{Coverage limit}$; the return period is $1/ROL$.
- B. $ROL = \text{Coverage limit} / XL \text{ premium}$; it measures how many times the limit is paid for by the premium.
- C. $ROL = XL \text{ premium} / \text{Retention}$; it indicates the proportion of the expected loss retained.
- D. $ROL = \text{Historical loss experience} / \text{Ceded premium}$; it is equivalent to the combined ratio in reinsurance.

Answer P-04.

Rate On Line (ROL)

Correct answer: A

$$ROL = \frac{XL \text{ premium}}{\text{Coverage limit } (L)}.$$

Example: Premium = 200 M and $L = 4,000$ M.

$$ROL = \frac{200}{4,000} = 0.05 = 5\%.$$

The implied return period is:

$$\frac{1}{ROL} = \frac{1}{0.05} = 20 \text{ years.}$$

The layer is “exhausted” on average every 20 years *if the ROL were the probability of a total loss*.

Practical interpretation: A ROL of 5% means that, in the absence of loading, the reinsurer expects to recover its coverage limit through 20 years of accumulated premiums. Catastrophe reinsurers require a ROL significantly higher than the expected loss to cover volatility and the cost of capital.

PML — Probable Maximum Loss

The 250-year PML (99.5th percentile) is the Solvency II regulatory benchmark for the catastrophe SCR. A “well-designed” XL layer provides:

$$\text{Limit} \geq \text{PML}_{250} - \text{retention}$$

to ensure protection of economic capital.

P-05.

A stop-loss reinsurance contract triggers coverage when the cedent's aggregate loss ratio exceeds a threshold. If the actual loss ratio is 95% and the contract covers from 80% to 120%, how much does the reinsurer pay (on a total premium of 1,000 M)?

Calculation

- A. 150 M
- B. 200 M
- C. 950 M
- D. 0 M

Answer P-05.

Payment Calculation Under a Stop-Loss Contract

Correct answer: A — 150 M

Stop-loss covers the aggregate loss ratio. Coverage runs from 80% to 120% of earned premium.

Expressing amounts in millions of COP:

$$\begin{aligned}\text{Total premium} &= 1,000, \\ \text{Losses} &= 95\% \times 1,000 = 950, \\ \text{Lower threshold (attachment)} &= 80\% \times 1,000 = 800, \\ \text{Upper limit (exhaustion)} &= 120\% \times 1,000 = 1,200.\end{aligned}$$

The losses fall within the layer, since $950 > 800$ and $950 < 1,200$. Therefore:

$$\begin{aligned}\text{Payment}_{\text{reins}} &= \min(\text{Losses} - \text{Threshold}, \text{Layer limit}) \\ &= \min(950 - 800, 1,200 - 800) \\ &= \min(150, 400) \\ &= 150 \text{ million COP.}\end{aligned}$$

If the loss ratio had been 130% (1,300 M), the reinsurer would pay the maximum layer limit:

$$1,200 - 800 = 400 \text{ M.}$$

The cedent absorbs the additional 100 M above the exhaustion point.

Typical Use of Stop-Loss

Protection of underwriting results in portfolios with highly volatile frequency and aggregate severity (SOAT, agricultural hail). In Colombia, it is widely used in personal insurance and group health reinsurance to control loss ratios in contracts with corporate clients.

P-06.

Which of the following is the most technically appropriate approach to setting the premium for a catastrophe XL contract?

Methodology

- A. Calculate the layer's historical burning cost and apply a safety loading factor.
- B. Use only the Lloyd's curve without considering the local exposure profile.
- C. Combine trend-adjusted burning cost with catastrophe simulation models (CAT models) and analyze the PML at different return periods.
- D. Apply the market ROL directly without analyzing the company's own exposure.

Answer P-06.

Technical Approach to Catastrophe XL Pricing

Correct answer: C

Pricing catastrophe layers requires a combination of:

1. **Adjusted Burning Cost:** historical layer losses divided by earned premium, adjusted for trend and inflation. Limitation: catastrophe events are scarce; burning cost is only valid for frequent events.

$$BC_{\text{layer}} = \frac{\sum_i \min(\max(S_i - M, 0), L)}{\text{Earned premium for the period}}.$$

$$BC_{\text{adjusted}} = BC_{\text{raw}} \times \text{Trend factor} \times \text{Exposure factor}.$$

2. **CAT models (RMS, AIR, CoreLogic):** probabilistic models of events, vulnerability, and exposure. They generate loss distributions by layer and allow the calculation of:

OEP(T): Loss exceeded with probability $1/T$ (per occurrence).

AEP(T): Annual aggregate loss exceeded with probability $1/T$.

$$\text{PML}_{250} = \text{OEP}(250 \text{ years}).$$

3. **Market analysis (reference ROL):** serves as a benchmark to check whether the technical price is competitive, not as a substitute for the company's own analysis.

Hannover Re and CAT Models

Hannover Re has proprietary access to its own CAT models and makes them available to the group's cedents to structure XL programs. As an HDI actuary, you are likely to be asked how a CAT model is calibrated for the Colombian portfolio (Zone II earthquakes, rainfall in Bogotá/Medellín).

P-07.

What are the two main advantages of facultative reinsurance over treaty reinsurance?

Conceptual

- A. Greater automaticity and lower transaction costs; it covers the entire portfolio without exception.
- B. It allows individual risks to be selected and provides additional capacity for risks exceeding the automatic treaty limits.
- C. It is cheaper than treaty reinsurance and does not require the reinsurer's approval for each risk.
- D. It guarantees fixed pricing terms throughout the term of the framework agreement.

Answer P-07.

Facultative Reinsurance: Specific Advantages

Correct answer: B

Facultative reinsurance is negotiated risk by risk (or policy by policy), unlike treaty reinsurance, which automatically covers the entire defined portfolio.

Advantage 1 — Individual Selection: The cedent can cede only the largest or specific risks without requiring the reinsurer to take the entire portfolio. The reinsurer, in turn, can accept or reject each risk with full information.

Advantage 2 — Additional Capacity: When an individual risk exceeds the treaty's automatic limits (e.g., an industrial plant with an SI of 500,000 M), facultative reinsurance provides the incremental capacity.

Example: The treaty covers a maximum of 10,000 M per risk. For risk A with an SI of 30,000 M:

$$\begin{aligned}\text{Treaty coverage} &= 10,000 \text{ M}, \\ \text{Facultative capacity} &= 30,000 - 10,000 \\ &= 20,000 \text{ M}.\end{aligned}$$

Facultative reinsurance is placed for the remaining 20,000 M and negotiated directly with one or more reinsurers.

Disadvantage of Facultative Reinsurance

Higher transaction costs (individual negotiation), less automaticity, and no placement guarantee. At HDI Colombia, it is mainly used for engineering risks, large industrial risks, and liability policies with high sums insured, consistent with the Talanx Group's Mittelstand DNA.